

RIK JANSEN

Senior Software Engineer

rik.jansen.ict@outlook.com | linkedin.com/in/rik-jansen-b24a463a1/ | Roden, Drenthe, Netherlands | +31 685549850

SUMMARY

Python Cloud Engineer with 11+ years of experience across pharmaceutical R&D, SaaS, healthcare, and scientific workflow platforms, delivering AWS cloud-native systems with **Python 3.8-3.12**, **FastAPI 0.95-0.110**, **Terraform 1.3-1.7**, **Docker 20.10-24.x**, **Kubernetes 1.24-1.29**, and **Revvity Signals VitroVivo 3.0-3.3**. Specialized in PaaS/SaaS architecture, secure APIs, CI/CD automation, observability, disaster recovery, and lab informatics workflows, with delivery experience in regulated environments where reliability, reproducibility, and data traceability are critical. Known for improving release quality and turning complex research workflows into scalable cloud services.

SKILLS

- **Languages:** Python 3.8-3.12, TypeScript 4.x-5.x, JavaScript ES6+, Bash, SQL
- **Cloud Platforms:** AWS, ECS, EKS, Lambda, API Gateway, RDS, S3, CloudFront, Route 53, IAM, VPC, CloudWatch
- **Backend Engineering:** FastAPI 0.95-0.110, Flask 1.x-2.x, Django 3.x-4.x, REST APIs, GraphQL, WebSockets, async services
- **Infrastructure & DevOps:** Terraform 1.3-1.7, AWS CloudFormation, Docker 20.10-24.x, Kubernetes 1.24-1.29, Helm 3.x
- **CI/CD:** GitHub Actions, GitLab CI, Jenkins, Azure DevOps, Newman, Postman, automated rollback, blue-green delivery
- **Scientific & Pharma Workflows:** Revvity Signals VitroVivo 3.0-3.3, Signals Notebook integration, assay workflows, in vitro/in vivo data pipelines
- **Architecture:** PaaS/SaaS, multi-tenancy, API Gateway patterns, event-driven systems, service mesh,DDD, Clean Architecture
- **Observability:** CloudWatch, OpenTelemetry 1.x, Prometheus 2.x, Grafana 9.x-10.x, structured logging, SLO/SLA monitoring
- **Data & Messaging:** PostgreSQL 12-16, MySQL 8.x, Redis 6.x-7.x, RabbitMQ 3.x, SQS, SNS, EventBridge
- **Security:** IAM least privilege, secrets management, KMS, secure provisioning, vulnerability scanning, SDLC security reviews

EXPERIENCE

Senior Software Engineer

Viario Networks Inc.

June 2021 - October 2025

- Architected **Python 3.9-3.12** cloud-native microservices on **AWS EKS, ECS, Lambda, API Gateway, RDS, S3, and CloudWatch** for pharma-facing SaaS platforms supporting regulated scientific workflows.
- Designed scalable **PaaS/SaaS architecture** with multi-tenant isolation, tenant-aware API routing, shared service boundaries, and secure provisioning patterns for enterprise research teams.
- Implemented **Revvity Signals VitroVivo 3.0-3.3** workflow integrations for assay data processing, study templates, Notebook-linked exports, and controlled in vitro/in vivo analysis pipelines.
- Built **FastAPI 0.95-0.110** services using async workers, Pydantic validation, OpenAPI contracts, and structured error handling to reduce API defects by **23%** across release cycles.
- Migrated legacy Flask services into modular **FastAPI** applications with cleaner domain boundaries, improved test coverage, and safer deployment paths without disrupting active scientific users.
- Created reusable **Terraform 1.3-1.7** modules for VPC, EKS, RDS, IAM, KMS, S3, and CloudWatch resources, reducing environment provisioning effort by **28%**.
- Led Kubernetes improvements across **Kubernetes 1.24-1.29**, including ingress hardening, HPA tuning, pod disruption budgets, secrets handling, and safer rollout strategies.
- Built GitHub Actions and Jenkins pipelines with automated tests, Postman/Newman API validation, image scanning, staged deployments, and rollback controls for production releases.
- Designed AWS disaster recovery plans with defined **RPO/RTO**, cross-region backup validation, restore testing, Route 53 failover checks, and controlled chaos testing scenarios.
- Implemented **OpenTelemetry 1.x**, Prometheus, Grafana, and CloudWatch dashboards to improve production visibility and reduce mean time to resolution by **19%**.
- Resolved a recurring VitroVivo data-sync issue by tracing inconsistent payload mappings between Notebook exports and downstream pipelines, then adding schema validation and replay-safe workers.
- Collaborated with product, QA, data science, and platform teams to introduce practical **Generative AI** workflow assistants for documentation, API testing, and scientific data review support.

Software Engineer

ArnAmy, Inc.

August 2017 - May 2021

- Developed **Python 3.7-3.10** backend services for healthcare and SaaS workflow platforms, focusing on secure APIs, deployment automation, and reliable data processing.
- Built REST and GraphQL APIs with **Django 3.x-4.x**, Flask 1.x-2.x, PostgreSQL, Redis, and Celery for high-volume operational workflows and reporting features.
- Implemented AWS services including **ECS, Lambda, API Gateway, S3, RDS, IAM, SQS, SNS, and CloudWatch** to support scalable application workloads.
- Introduced infrastructure automation with **Terraform 0.14-1.3** and CloudFormation, reducing manual cloud configuration errors by **22%**.
- Containerized backend applications with **Docker 19.x-20.10** and supported early Kubernetes workloads using Helm charts, ingress rules, config maps, and secret references.
- Improved CI/CD pipelines in GitLab CI and Jenkins with unit tests, integration tests, container builds, environment promotion, and rollback procedures.
- Designed API test suites with **Postman** and **Newman**, allowing regression tests to run automatically before staging and production deployments.
- Resolved production timeout issues by profiling slow SQL joins, adding targeted indexes, optimizing pagination, and introducing Redis caching for repeated queries.
- Migrated older Python services from **Python 3.6** patterns toward **Python 3.9-3.10**, replacing deprecated libraries and improving runtime consistency.
- Added observability with CloudWatch metrics, structured logs, alert rules, and Grafana dashboards, improving incident detection speed by **17%**.
- Worked closely with QA, DevOps, and product owners to deliver secure workflow features while keeping release documentation and operational runbooks accurate.

Software Developer

Webiz Digital

December 2015 - July 2017

- Built and maintained web backends that exposed consistent REST endpoints, focusing on predictable error handling, input validation, and maintainable service structure for long-term evolution.
- Implemented integration points with external services using well-defined adapters, adding retries, timeouts, and logging to keep third-party dependencies from destabilizing core application flows.
- Improved application performance by introducing caching strategies and optimizing database queries, reducing page load times and increasing reliability during traffic spikes and campaign peaks.
- Developed automation for build and release steps to shorten feedback loops, ensuring changes were tested early and deployments were repeatable across environments with fewer manual steps.
- Enhanced operational visibility by adding health endpoints and structured logs, enabling faster diagnosis when issues appeared and reducing downtime through quicker troubleshooting cycles.
- Collaborated with designers and stakeholders to implement responsive interfaces and reliable data flows, ensuring the system behaved consistently across browsers and common user journeys.
- Hardened security fundamentals through consistent sanitization and safer session handling, reducing exposure to common web vulnerabilities while keeping developer ergonomics practical.
- Refactored legacy areas into clearer modules with stable interfaces, improving readability and making it easier for the team to change individual parts without unintended side effects.
- Supported incident resolution by reproducing production issues locally, identifying root causes, and delivering targeted fixes that improved stability rather than masking underlying problems.

Junior Software Engineer

High Touch Technologies

August 2014 - November 2015

- Built backend and cloud-supported web applications using **Python 3.5-3.7**, Django 1.11-2.2, Flask, PostgreSQL, MySQL, Redis, and JavaScript.
- Developed secure REST APIs for SaaS and digital workflow platforms, applying authentication, authorization, input validation, and reusable service-layer patterns.
- Supported AWS deployments using **EC2, S3, RDS, Elastic Beanstalk, IAM, Route 53, and CloudWatch** for small and mid-sized production systems.

- Created Docker-based development environments with **Docker 17.x-19.x**, reducing setup inconsistencies and helping developers reproduce bugs faster.
- Improved deployment scripts and Jenkins jobs for automated builds, database migrations, static asset handling, and controlled release promotion.
- Migrated selected legacy Python 2.7 components to **Python 3.6**, fixing dependency conflicts, Unicode handling issues, and outdated package usage.
- Resolved recurring API 500 errors by improving exception handling, request validation, logging coverage, and database transaction safety.
- Optimized slow reporting endpoints through query refactoring, indexes, and cached aggregates, improving response times by **21%**.
- Implemented WebSocket-based status updates for operational dashboards where users needed live feedback during long-running background tasks.
- Collaborated with frontend engineers and client stakeholders to convert business requirements into maintainable backend services and deployment-ready features.

EDUCATION

Bachelor's degree in Computer Science

University of Amsterdam

2010 - 2014